

TECHNICAL BLUEPRINT · PROBLEM 4/22

Loaded Truss — *Method of Joints*

An interactive diagram-driven solver for all 13 members of a loaded truss from Meriam & Kraige — Engineering Mechanics: Statics

REFERENCE

Meriam & Kraige, Problem 4/22

TECH STACK

HTML5 · CSS3 · JavaScript · Canvas

MEMBERS SOLVED

All 13 (complete truss)

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§ 1 PROBLEM STATEMENT

Meriam & Kraige Problem 4/22 — A loaded Pratt truss has five top-chord nodes **A, B, C, D, E** spaced at equal horizontal distance d , and three bottom-chord nodes **H, G, F** hanging at depth d below the top chord.

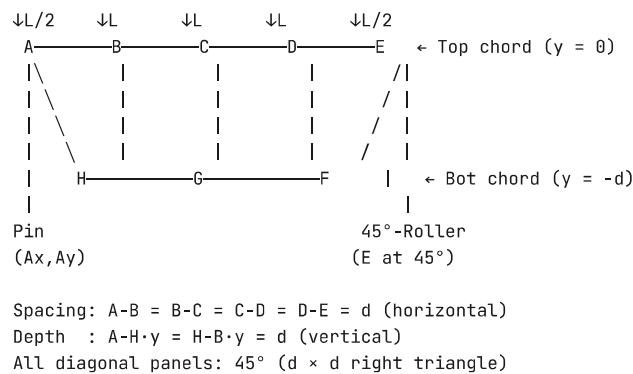
Applied loads act downward: $L/2$ at nodes **A** and **E**, and L at nodes **B, C, D**. Support conditions: a **pin** at **A** and a **45°-inclined roller** at **E**.

The student must determine the forces in **all 13 members** of the truss, classifying each as Tension (T), Compression (C), or zero-force.

CORE CONCEPT

This problem applies the **Method of Joints** — at each pin joint, the two equilibrium equations ($\Sigma F_x = 0$ and $\Sigma F_y = 0$) are used to find unknown member forces. Joints are solved in a sequence beginning with those having only two unknowns. Member **CG = 0** is identified as a zero-force member from joint G inspection.

§ 2 TRUSS GEOMETRY & LOADING



Symbol	Description	Default Value	Unit
d	Panel spacing (horizontal & vertical)	1 000	mm
L	Full panel load applied at B, C, D	10 000	N
$L/2$	Half-panel load applied at A and E	5 000	N
A_x, A_y	Pin reactions at A	2L, 2L	N
E	Roller reaction at E (45° direction)	$2\sqrt{2} \cdot L$	N

§ 3 MEMBER INVENTORY (13 MEMBERS)

Group	Members	Count	Angle
Top chord	AB, BC, CD, DE	4	0° (horizontal)
Bottom chord	HG, GF	2	0° (horizontal)
Verticals	BH, CG, DF	3	90° (vertical)
Diagonals	AH, HC, CF, FE	4	45°
Total		13	

§ 4 GLOBAL EQUILIBRIUM — REACTIONS

Moment about A ($\Sigma M_A = 0$)

Taking clockwise moments as positive, with the roller reaction E perpendicular to the 45° surface (i.e., at 45° to horizontal, components $E \cdot \cos 45^\circ$ leftward and $E \cdot \sin 45^\circ$ upward):

$$-L \cdot d - L \cdot 2d - L \cdot 3d - (L/2) \cdot 4d + E \cdot (1/\sqrt{2}) \cdot 4d = 0$$

$$-8Ld + 2\sqrt{2} \cdot E \cdot d = 0 \implies E = 2\sqrt{2} \cdot L$$

Force Equilibrium

$$\Sigma F_x = 0: A_x = E \cdot \cos 45^\circ = 2L$$

$$\Sigma F_y = 0: A_y + E \cdot \sin 45^\circ - 4L = 0 \implies A_y = 2L$$

SYMMETRY CHECK

By the symmetry of the loading ($L/2$ at both ends, L at all interior nodes) and geometry, $A_y = E \cdot \sin 45^\circ = 2L$ — the truss is loaded symmetrically, confirming equal vertical reactions.

§ 5 METHOD OF JOINTS — DERIVATION**Joint A → Members AH (T) and AB (C)**

External at A: $A_x = 2L$ ($\rightarrow$), $A_y = 2L$ ($\uparrow$), applied load = $L/2$ ($\downarrow$). Members: AB (horizontal $\rightarrow$) and AH (diagonal 45° down-right).

$$\Sigma F_y = 0: 2L - L/2 - AH/\sqrt{2} = 0 \Rightarrow AH = 3\sqrt{2}L/2 \text{ (T)}$$

$$\Sigma F_x = 0: 2L + AH/\sqrt{2} - AB = 0 \Rightarrow AB = 7L/2 \text{ (C)}$$

Joint E → Members FE (T) and DE (C)

By symmetry with joint A: $FE = 3\sqrt{2}L/2$ (T) and $DE = 7L/2$ (C).

Joint B → Members BH (C) and BC (C)

External at B: load = L ($\downarrow$). Members: AB, BC (horizontal), BH (vertical $\downarrow$).

$$\Sigma F_y = 0: -L + BH \cdot (-1) = 0 \Rightarrow BH = L \text{ (C)}$$

$$\Sigma F_x = 0: BC = AB \Rightarrow BC = 7L/2 \text{ (C)}$$

Joint D → Members DF (C) and CD (C)

By symmetry with joint B: $DF = L(C)$ and $CD = 7L/2(C)$.

§ 6 JOINTS H, F AND G

Joint H → Members HC (C) and HG (T)

Known: $AH = 3\sqrt{2}L/2$ (T), $BH = L$ (C). Directions from H: to A = $(-1/\sqrt{2}, +1/\sqrt{2})$, to B = $(0, +1)$, to C = $(+1/\sqrt{2}, +1/\sqrt{2})$, to G = $(+1, 0)$.

$$\Sigma F_y: 3L/2 - L + HC/\sqrt{2} = 0 \Rightarrow HC = L\sqrt{2}/2 \text{ (C)}$$

$$\Sigma F_x: -3L/2 - L/2 + HG = 0 \Rightarrow HG = 2L \text{ (T)}$$

Joint F → Members CF (C) and GF (T)

By symmetry with joint H: $CF = L\sqrt{2}/2$ (C) and $GF = 2L$ (T).

Joint G → CG is a Zero-Force Member

At joint G, members HG (pulls left, T) and GF (pulls right, T) are equal and opposite. Member CG is vertical. With no external load at G:

$$\Sigma F_x = 0: -2L + 2L = 0 \checkmark$$

$$\Sigma F_y = 0: CG = 0 \Rightarrow CG = 0 \text{ (zero-force)}$$

VERIFICATION AT JOINT C

The diagonal members HC and CF each supply an upward vertical component of $L/\sqrt{2} \times 1/\sqrt{2} = L/2$. Together, $L/2 + L/2 = L$, which exactly balances the applied downward load L at C. ✓ The zero value of CG is confirmed.

§ 7 **COMPLETE RESULTS (DEFAULT: L = 10 kN, D = 1 M)**

TOP CHORD (COMPRESSION)

AB 35.00 kN COMPRESSION	BC 35.00 kN COMPRESSION	CD 35.00 kN COMPRESSION	DE 35.00 kN COMPRESSION
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BOTTOM CHORD (TENSION)

HG 20.00 kN TENSION	GF 20.00 kN TENSION
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VERTICALS

BH 10.00 kN COMPRESSION	CG 0 ZERO-FORCE	DF 10.00 kN COMPRESSION
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DIAGONALS

AH 21.21 kN TENSION	HC 7.07 kN COMPRESSION	CF 7.07 kN COMPRESSION	FE 21.21 kN TENSION
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§ 8 **SAMPLE CALCULATION (L = 10 000 N, D = 1 000 MM)**

Step 1 – Global equilibrium

$\Sigma M_A = 0: -Ld - L(2d) - L(3d) - (L/2)(4d) + E \cdot (1/\sqrt{2}) \cdot 4d = 0$
 $-8Ld + 2\sqrt{2} \cdot E \cdot d = 0$
 $E = 2\sqrt{2} \times 10000 = 28284.3 \text{ N}$
 $A_x = E \cdot \cos 45^\circ = 20000 \text{ N}$
 $A_y = 4L - E \cdot \sin 45^\circ = 20000 \text{ N}$

Step 2 – Joint A

$\Sigma F_y: 20000 - 5000 - AH/\sqrt{2} = 0$
 $AH = 15000 \times \sqrt{2} = 21213.2 \text{ N (T)}$
 $\Sigma F_x: 20000 + 21213.2/\sqrt{2} - AB = 0$
 $AB = 20000 + 15000 = 35000 \text{ N (C)}$

Step 3 – Joint B

$\Sigma F_y: -10000 - BH = 0 \rightarrow BH = 10000 \text{ N (C)}$
 $\Sigma F_x: BC = AB \rightarrow BC = 35000 \text{ N (C)}$

Step 4 – Joint H

$\Sigma F_y: 15000 - 10000 + HC/\sqrt{2} = 0 \rightarrow HC = -7071.1 \text{ N} \rightarrow HC = 7071.1 \text{ N (C)}$
 $\Sigma F_x: -15000 - 7071.1/\sqrt{2} + HG = 0 \rightarrow HG = 20000 \text{ N (T)}$

Step 5 – Joint G

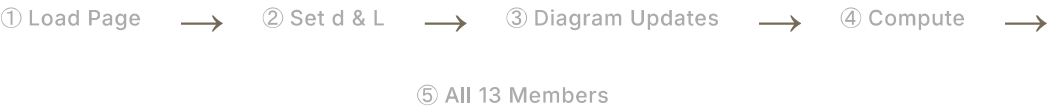
$\Sigma F_y: CG = 0 \rightarrow CG = 0 \text{ (zero-force)}$

Steps 6,7,8 – By symmetry (D, F, E) give CD, DF, GF, CF, FE, DE.

Top chord	AB=BC=CD=DE	: 35.00 kN (C)
Bot chord	HG=GF	: 20.00 kN (T)
Verticals	BH=DF	: 10.00 kN (C)
Vertical	CG	: 0 (zero-force)
Diag outer	AH=FE	: 21.21 kN (T)
Diag inner	HC=CF	: 7.07 kN (C)

§ 9 **APPLICATION USER FLOW**

The interactive solver follows a five-step workflow:



- **Load Page:** The solver loads with default values ($d = 1\,000\text{ mm}$, $L = 10\,000\text{ N}$) and renders the live canvas truss diagram with grid, dimension lines, and load arrows.
- **Set d & L :** Two input fields allow the student to parametrically change the panel spacing and applied load. Symbolic inputs (e.g., " L ") are supported.
- **Diagram Updates:** The HTML5 Canvas redraws in real time as values change — nodes reposition, members rescale, load arrows update.
- **Compute:** Clicking "Compute Solution" runs the full method-of-joints solver for all 13 members in sequence.
- **All 13 Members:** Results appear in four grouped sections (Top Chord, Bottom Chord, Verticals, Diagonals), color-coded: **cyan** = Compression, **green** = Tension, **grey** = Zero-force.

ZERO-FORCE MEMBER DETECTION

Member CG is identified analytically (not by inspection of only horizontal members) by solving joint G after HG and GF are determined. Since $HG = GF = 2L$, they cancel in $\sum F_x$, leaving $\sum F_y$: $CG = 0$. The solver correctly identifies this — an important correction to the textbook's "by inspection" shortcut which fails when diagonal members (HC, CF) are also present at joint C.

§ 10 UI DESIGN & LAYOUT

Screen Layout

The application uses the same two-column layout as Problem 2/50:

- **Left panel ($\approx 70\%$):** An HTML5 `<canvas>` showing the truss with colored members, node dots, labelled nodes, load arrows, dimension lines, support symbols, reaction labels, and a legend.
- **Right panel ($\approx 30\%$):** Two input fields (spacing d and load L) plus a read-only info block describing support conditions.

Color Coding

Color	Meaning
Blue (#4a90d9)	Truss members (canvas diagram)
Red (#ff2255)	Applied downward loads
Green (#39ff14)	Dimension annotations
Cyan (#00e5ff)	Support reactions (A_x , A_y , E)
Cyan (result)	Compression member values
Green (result)	Tension member values
Muted grey	Zero-force member (CG)

§ 11 CODE ARCHITECTURE

```
// Key functions in problem_4_22.html

resizeCanvas()      // DPR-aware canvas resize
drawDiagram()       // Full redraw: grid, members, nodes, labels,
                    // arrows, dims, supports, reactions, legend
drawArrow(...)      // Generic arrow with label
drawDimLine(...)    // Horizontal dimension annotation
drawDimLineV(...)   // Vertical dimension annotation
drawPin(...)        // Pin support symbol at A
```

```

drawRoller45(...)    // 45°-inclined roller symbol at E
drawReactionLabels() // Post-compute: Ax, Ay, E values on canvas
drawLegend()         // Color-coded legend (top-right)

compute()             // Full 13-member solver (method of joints):
                      //   Reactions → A → E → B → D → H → F → G
                      //   Updates all ans-box, note, box class
                      //   Generates 9-step working HTML

toggleFullscreen()   // Canvas fullscreen mode (Esc to exit)
parseField(id, def)  // Numeric/symbolic input parser

```

DESIGN DECISIONS

- The truss diagram uses **auto-scaling** — the `unitPx` variable adapts so the truss always fits within available canvas space regardless of window size.
- Result boxes are dynamically re-classed (e.g., `ans-box compression`) so their CSS color updates automatically on compute.
- Steps are rendered as HTML strings with `<span>` color tags, supporting both numeric and symbolic output.

Problem 4/22 Technical Blueprint · Rathanga Parthasarathy · March 2026
 Meriam & Kraige — Engineering Mechanics: Statics · Method of Joints · All 13 Members